Checkpoint 3.

$\min z = 3y_1 + 4y_2 + 5y_3 \quad c_1 = 3 \quad c_2 = 4 \quad c_3 = 5$

a) $\pi = (1, 2, 3)$ b) $\pi' = (2, 2, 1)$

T_1: $y_1 + y_2 \geq 1$

T_2: $y_2 + y_3 \geq 1$

T_3: $y_1 + y_3 \geq 1$

$0 \leq y_i \leq 1$

$\pi_1 = 1$

a) $\begin{cases} y_1 + y_2 = 1 + \varepsilon \\ y_2 + y_3 = 1 \\ y_1 + y_3 = 1 \end{cases} \Rightarrow \begin{array}{l} y_1 - y_3 = \varepsilon \\ + \; y_1 + y_3 = 1 \end{array}$

$= 2y_1 + 1 + \varepsilon$

$y_1 = \frac{1+\varepsilon}{2}$

Substitute $y_1 + y_3 = 1$, $y_3 = 1 - \frac{1+\varepsilon}{2} = \frac{1-\varepsilon}{2}$

From $y_2 + y_3 = 1$,

$y_2 = 1 - \frac{1-\varepsilon}{2} = \frac{1+\varepsilon}{2}$

$y^*(\varepsilon) = \left(\frac{1+\varepsilon}{2}, \frac{1+\varepsilon}{2}, \frac{1-\varepsilon}{2}\right)$

Substitute into $\min z = 3y_1 + 4y_2 + 5y_3$:

$z(1+\varepsilon) = 3\left(\frac{1+\varepsilon}{2}\right) + 4\left(\frac{1+\varepsilon}{2}\right) + 5\left(\frac{1-\varepsilon}{2}\right) = \frac{3+4+5+(3+4-5)\varepsilon}{2}$

$= \frac{12 + 2\varepsilon}{2} = 6 + \varepsilon \Rightarrow$ The slope with respect to RHS is $1 = \pi_1$

b) $0 \le y_i \le 1$

$\frac{1+\varepsilon}{2} \le 1 \Rightarrow \varepsilon \le 1$, $\frac{1-\varepsilon}{2} \ge 0 \Rightarrow \varepsilon \le 1$

At $\varepsilon = 1$ we get $(1,1,0)$ which satisfies constraints.

So valid for $\underline{0 \le \varepsilon \le 1}$.

c) Interpretation.

If T_1 means 1st customer must be covered, then $\bar{a}_1 = 1$ means increasing the coverage required by one unit also raises the objective value by one unit (locally), and the current optimal basis remains valid.